

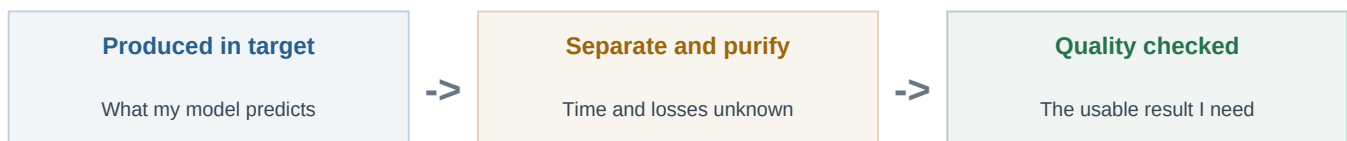
I can predict what is produced. I cannot yet predict what becomes usable.

The gap between those two numbers may matter more than another small accuracy improvement.

I am a 12th grader at Glen Burnie High School, and I built a reduced model of the Ra-226 to Ac-225 production pathway. My equations follow irradiation, isotope buildup, decay, and cooling. They stop at the exact point where isotope harvesting and chemistry begin, which is why I am reaching out to your group.

THE RESULT THAT CHANGED HOW I SEE THE PROJECT

When I reconstructed the published Berkeley experiments, I had to keep produced activity and recovered activity as different quantities. The recovery was not identical between the two experiments. That made something obvious: a schedule that produces the most atoms is not automatically the schedule that delivers the most usable Ac-225, or does it fastest.



The practical numbers I am missing

- Time from end of irradiation to completed separation.
- Recovery fraction and how it should be defined.
- Losses during purification, transfers, and quality checks.
- The point when material is considered usable rather than merely produced.

What I can do without retraining V3

- Add a separate chemistry and recovery layer after the frozen physics model.
- Run sensitivity ranges instead of pretending one facility value is universal.
- Rank usable activity and total turnaround, then verify the nuclear part with the trusted solver.
- Clearly label every value as published, estimated, or still unknown.

What I am asking: Could you recommend one public isotope-harvesting or radiochemistry process example that reports separation time, recovery, purification losses, and quality-control steps? If a person in your group works more directly with those measurements, I would be grateful for the right name.

PERSONAL NOTE FOR

Professor Gregory W. Severin

Michigan State University and FRIB Isotope Harvesting Group